

TEMPORAL MINING

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# STEP 1: Load Dataset
df = pd.read_csv('air_quality_dataset.csv')
print("Columns:", df.columns)

# STEP 2: Use PM2.5 as Value
df = df[['PM2.5']].dropna()
df.rename(columns={'PM2.5': 'Value'}, inplace=True)

# STEP 3: Create Time Index
df['Time'] = range(len(df))
df.set_index('Time', inplace=True)

# STEP 4: Plot Trend
plt.figure()
plt.plot(df.index, df['Value'])
plt.title("Air Quality Trend (PM2.5)")
plt.xlabel("Time")
plt.ylabel("PM2.5")
plt.show()

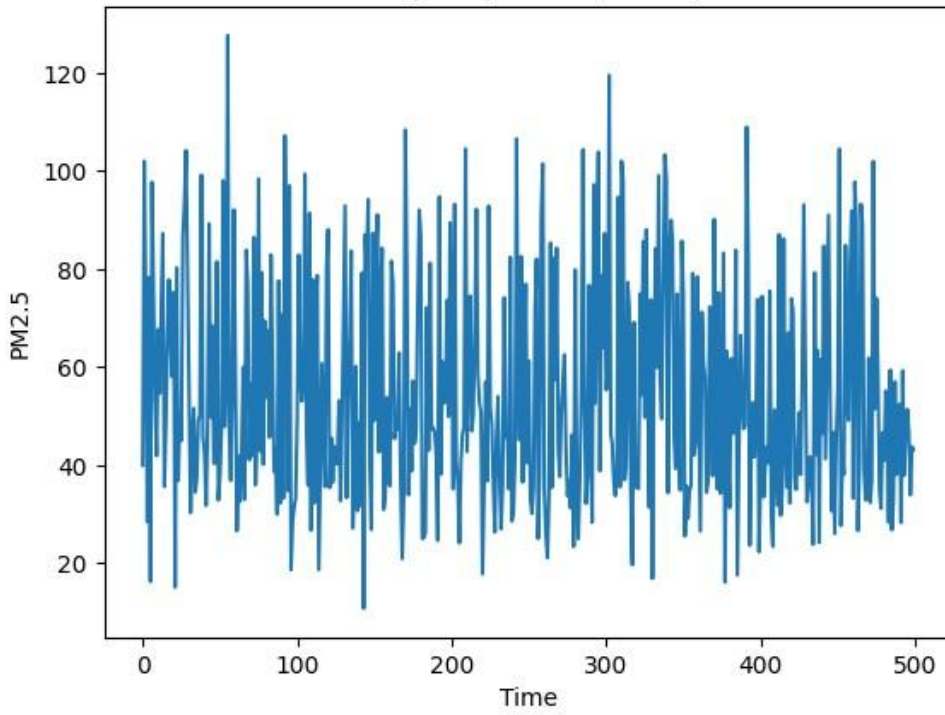
# STEP 5: Moving Average
df['Moving_Avg'] = df['Value'].rolling(window=20).mean()
plt.figure()
plt.plot(df.index, df['Value'], alpha=0.3)  # fade original
plt.plot(df.index, df['Moving_Avg'], linewidth=2)
plt.title("Smoothed Air Quality Trend")
plt.xlabel("Time")
plt.ylabel("PM2.5")
plt.show()

# STEP 6: Clustering
kmeans = KMeans(n_clusters=3, random_state=0, n_init=10)
df['Cluster'] = kmeans.fit_predict(df[['Value']])
plt.figure()
plt.scatter(df.index, df['Value'], c=df['Cluster'])
plt.title("Air Quality Clustering")
plt.xlabel("Time")
plt.ylabel("PM2.5")
plt.show()

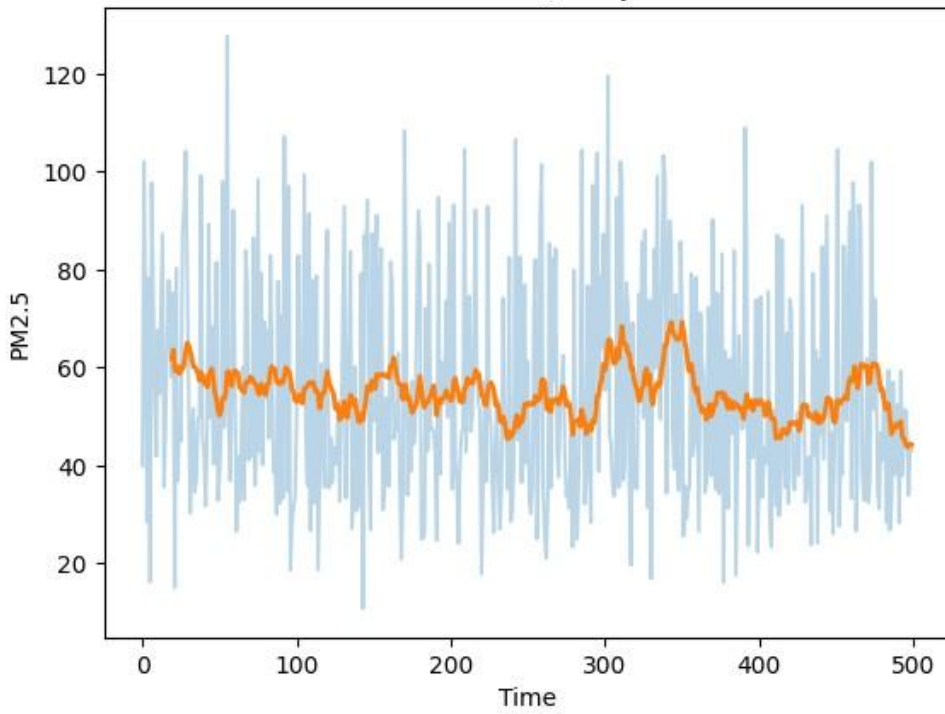
# STEP 7: Output
print(df.head())
```

OUTPUT:

Air Quality Trend (PM2.5)



Smoothed Air Quality Trend



Air Quality Clustering

